Asssignment 6

Chapter 6

1. Show that $(\mathbf{x}, \mathbf{y}) \in T^n_{[X,Y]\delta}$ and $(\mathbf{y}, \mathbf{z}) \in T^n_{[Y,Z]\delta}$ do not imply $(\mathbf{x}, \mathbf{z}) \in T^n_{[X,Z]\delta}$.

2. Let $\mathbf{X} = (X_1, X_2, \cdots, X_n)$, where X_k are i.i.d. with generic random variable X. Prove that

$$\Pr\{\mathbf{X} \in T^n_{[X]\delta}\} \geq 1 - \frac{|\mathcal{X}|^3}{n\delta^2}$$

   for any n and $\delta > 0$. This shows that $\Pr\{\mathbf{X} \in T^n_{[X]\delta}\} \to 1$ as $\delta \to 0$ and $n \to \infty$ if $\sqrt{n}\delta \to \infty$.

4. Prove Proposition 6.13. Hint: Use the fact that if $(\mathbf{X}, \mathbf{Y}) \in T^n_{[XY]\delta}$, then $\mathbf{X} \in S^n_{[X]\delta}$.

6. Let p be any probability distribution over a finite set $\mathcal{X}$ and η be a real number in $(0, 1)$. Prove that for any subset A of $\mathcal{X}^n$ with $p^n(A) \geq \eta$,

$$|A \cap T^n_{[X]\delta}| \geq 2^{n(H(p)-\delta')},$$

   where $\delta' \to 0$ as $\delta \to 0$ and $n \to \infty$.

In the following problems, for a sequence $\mathbf{x} \in \mathcal{X}^n$, let $q_{\mathbf{x}}$ be the empirical distribution of $\mathbf{x}$, i.e., $q_{\mathbf{x}}(x) = n^{-1}N(x; \mathbf{x})$ for all $x \in \mathcal{X}$. Similarly, for a pair of sequences $(\mathbf{x}, \mathbf{y}) \in \mathcal{X}^n \times \mathcal{Y}^n$, let $q_{\mathbf{x},\mathbf{y}}$ be the joint empirical distribution of $(\mathbf{x}, \mathbf{y})$, i.e., $q_{\mathbf{x},\mathbf{y}}(x, y) = n^{-1}N(x, y; \mathbf{x}, \mathbf{y})$ for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$.

7. *Alternative definition of strong typicality* Show that (6.1) is equivalent to

$$V(q_{\mathbf{x}}, p) \leq \delta,$$

   where $V(\cdot, \cdot)$ denotes the variational distance. Thus strong typicality can be regarded as requiring the empirical distribution of a sequence to be close to the probability distribution of the generic random variable in variational distance. Also compare the result here with the alternative definition of weak typicality (Problem 5 in Chapter 5).

8. The empirical distribution $q_{\mathbf{x}}$ of the sequence $\mathbf{x}$ is also called the *type* of $\mathbf{x}$. Assuming that $\mathcal{X}$ is finite, show that there are a total of $\binom{n+|\mathcal{X}|-1}{n}$ distinct types $q_{\mathbf{x}}$. Hint: There are $\binom{a+b-1}{a}$ ways to distribute a identical balls in b boxes.